Flight Training Lesson Guide
CAT II/III Certification with PRM/PAR

Revisions

Any questions about the content or use of this manual can be directed to:

Fleet Captain - Atlas Air Training Center
Standardsreport747@atlasair.com

Revision Highlights

- Added notes for SMIGS
- Amended PRM guidance
- Guidance for Seat Dependent training

Lesson Outline

The lesson provides training and qualification in All Weather Operations (AWO), radar approaches (PAR), PRM/LDA, and CAT II / CAT III ILS procedures.

- AWO and CAT II / III training conducted in KSEA
- PRM/LDA and PAR training conducted in KSFO
- Normal procedures
- Normal CAT III approach to a full stop
- Engine Inoperative CAT-II approach to a missed approach
- Engine Inoperative CAT-II approach to a full stop
- Precision Runway Monitored (PRM) approach and breakout
- Radar approaches (PAR)
- DoD FLIP approach chart (briefing room only)

Grading

Demonstrate proficiency in AWO operations.

CAT II / CAT III certification must be accomplished by a Check Airman.

Document CAT II / CAT III certification on a PC form with a “S” along with grading the PRM, PAR, and Seat Specific (if required) with a “C”.

Objective

To train the skills and knowledge required to accomplish the maneuvers, procedures, and duties as captain or first officer.

Training

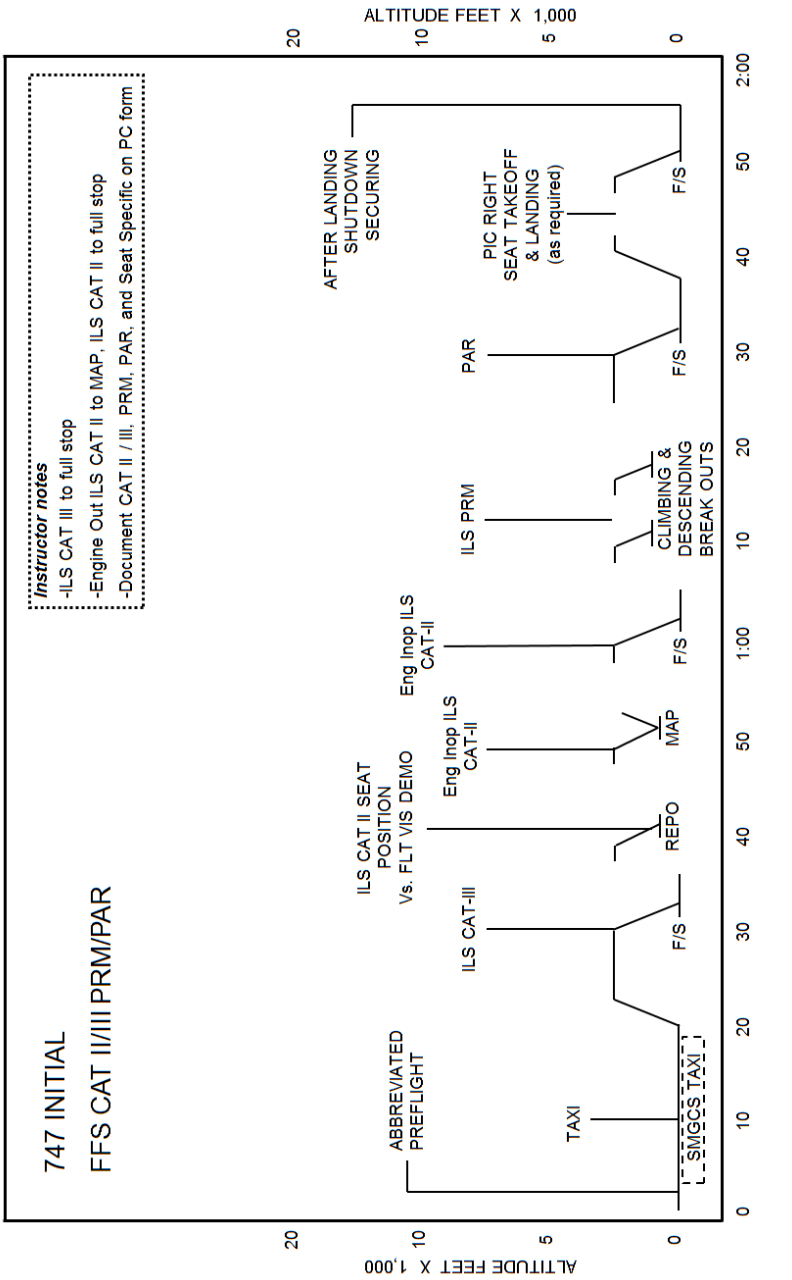
- Portable Electronic Flight Bag (pEFB)
- Glare-shield Checklists
- QRH Section AWO
- Flight Data Includes:
  - Flight Plan
  - DDG
  - Weather
  - Weight & Balance Data
  - Clearance
  - TLR / FDP

Briefing Items

- Review paperwork
- Normal Preflight
- SMGCS/LVP/Surface Incursion Prevention
  - ➢ Note: Some Airports now use 1200 to 500 RVR and 500 RVR and below charts
- Low visibility takeoff (500ft / 150m RVR)
  - Lowest visibility for a particular runway
- CAT II and CAT III Approaches
  - Briefing / callouts
  - Visibility / RVR requirements
  - CA and FO duties
  - Mode Failures
  - Course / Glidepath Deviations
- Rejected Takeoff
  - Low Visibility Considerations
- Engine Inoperative CAT II Approach
  - A/P and A/T
  - A/P, no A/T
- One Engine Inoperative Missed approach
- One Engine Inoperative CAT III Approach
  - Landing with Rollout mode failure
- PRM/LDA Approach
  - Flown by Captains
  - Have student review the Attention All User Page
    - PRM monitor frequency
    - Use glideslope
    - Report traffic in sight, do not pass. Must be acquired prior to MAP
    - Breakout phraseology – “TRAFFIC ALERT”
  - Review FCOM Breakout Maneuver

- PAR Approach
  - Review sample PAR charts for KNGU provided in this guide
  - Review PRM approach in maneuvers section of FCOM
  - Flown by Captain
  - Flown with A/P and A/T
  - Use heading select and V/S as roll and pitch modes.
  - Prior to beginning final descent, the aircraft should be level at the intermediate altitude and “ALT” displayed as the pitch mode FMA.
- If required, PIC right seat takeoffs and manual landings.
- Right seat takeoff and landing
- Application of CRM/TEM skills, proactive planning and setting bottom lines during normal and non-normal procedures.

CAT II/III PRM/PAR Lesson Profile



OFP DATA –

ORIGIN:	SEATTLE, (KSEA)
DESTINATION:	AMSTERDAM, (EHAM)
FLIGHT NUMBER:	GTI 400
ROUTE:	KSEA BANGR_ ARRIE DCT PAE DCT DIABO YNY YMM DCT 67N090W DCT TOPPA EHAM
RUNWAY:	16C
DEPARTURE:	BANGR#_
TAKEOFF ALT:	KPDX

RAMP FUEL…….. 96.4 **TAXI FUEL (INFO ONLY)**.....0.9
RESERVE……..... 8.9 **COST INDEX**…….... 28
FLIGHT LEVEL…….. 370 **DISTANCE**……..……4401
DDG: NONE

LOAD SHEET DATA KILOS (LBS)

ZFW……..200.0 **TOGW**…… 296.4
CG……..21%

WEATHER

Seatac International Information *"Alpha" XXxxz, Wind 290/08, Visibility ¼ sm FG, sky condition Overcast 001, Temperature 23, Dew Point 22, Altimeter 30.05, Departing Runways 16C, RVR 16C 500/500/500. On initial contact advise you have information Alpha"*

CLEARANCE

Clearance*, Cleared To Amsterdam Schipol airport, BANGR_ Departure, ARRIE, then as filed. Climb and maintain 5,000 expect FL370 one zero minutes after departure, squawk 1747. Expect Runway 16C for departure.*

TLR DATA STRIP Kg / N1 (B5)

---DRY RWY – PTOW – ENG A/I OFF CALM WIND---
–

RWY	MTOW	MT	N1	CONFIG	FL	V1	VR	V2
16C	2964	49	95.9	D-TO2 PACKS ON	10	139	148	157

```
KSEA                CF680C2BxF
FLT NO     RLS NO      TIME
400            1      1951Z
WIND KTS    OAT C       QNH
220/08         23     30.05
------------------------
GRWT    296.4  VREF30 155
FUEL     96.4
ZFW     200.0

REMARKS
NONE

KSEA    16C            9426
MTOW    296.4   SMRGN 1406
FLAPS   20        PACKS ON
ATM     49C          V1 140
THR     D-TO2        VR 149
N1      *95.9        V2 158
CG      21.0      TRIM 6.5

DT H161         ACCEL 1000
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***N1 may differ slightly due to differences between B1/B5 engines.**

Example PAR - KNGU

N2

RADAR MINS

17033

RADAR INSTRUMENT APPROACH MINIMUMS

NORFOLK NS (KNGU), (CHAMBERS FIELD), VA (Amdt 5, 17033 USN) ELEV **15**
RADAR - (E) 119.2x 120.2x 279.525 322.525 335.8 353.55

	RWY	GS/TCH/RPI	CAT	DH/ MDA-VIS	HAT/ HATh/ HAA	CEIL-VIS
PAR[1]	10[2 3]	3.0°/54/1024	ABCDE	**114**-¼	100	(100-¼)
	28[4 10]	3.0°/37/709	ABCDE	**311**-¾	297	(300-¾)
PAR W/O GS[1]	28[5]		ABCDE	**380**-⅞	366	(400-⅞)
	10[11]		AB	**500**-⅝	486	(500-⅝)
			CDE	**500**-1	486	(500-1)
ASR	28[6]		ABCDE	**460**-1	446	(500-1)
	10[9]		AB	**680**-¾	666	(700-¾)
			CDE	**680**-1½	666	(700-1½)
C CIR PAR W/O GS[7 8]	10		A	**500**-1	485	(500-1)
			B	**560**-1	545	(600-1)
			C	**560**-1½	545	(600-1½)
			D	**580**-2	565	(600-2)
			E	**700**-2½	685	(700-2½)
	28		A	**500**-1⅛	485	(500-1⅛)
			B	**560**-1⅛	545	(600-1⅛)
			C	**560**-1½	545	(600-1½)
			D	**580**-2	565	(600-2)
			E	**700**-2½	685	(700-2½)
C CIR ASR[7 8]	28		A	**500**-1¼	485	(500-1¼)
			B	**560**-1¼	545	(600-1¼)
			C	**560**-1½	545	(600-1½)
			D	**580**-2	565	(600-2)
			E	**700**-2½	685	(700-2½)
	10		AB	**680**-1¼	665	(700-1¼)
			C	**680**-1⅜	665	(700-1⅜)
			D	**680**-2	665	(700-2)
			E	**700**-2½	685	(700-2½)

27 APR 2017 to 25 MAY 2017

27 APR 2017 to 25 MAY 2017

[1]No-NOTAM MP Sun 1230-1630Z++.
[2]When ALS inop, increase vis CAT ABCDE to ½ mile.
[3]PAPI RRP/TCH 670/35 not coincident with PAR RPI/TCH 1024/54.
[4]When ALS inop, increase vis CAT ABCDE to 1 mile.
[5]When ALS inop, increase vis CAT ABCDE to 1⅛ miles.
[6]When ALS inop, increase vis CAT AB to 1¼ miles, CAT CDE to 1⅜ miles.
[7]Cir not auth south of Rwy 10-28.
[8]Norfolk NS Heliport lctd 1 NM NW, use caution in cir aprch.
[9]When ALS inop, increase vis CAT AB to 1¼ miles, CAT CDE to 1⅞ miles.
[10]CAUTION: Wheel Crossing Height Group 3: 17 ft, Group 4: 12 ft is less than 20 ft.
[11]When ALS inop, increase vis CAT AB to 1 mile, CAT CDE to 1⅜ miles.

NE-3

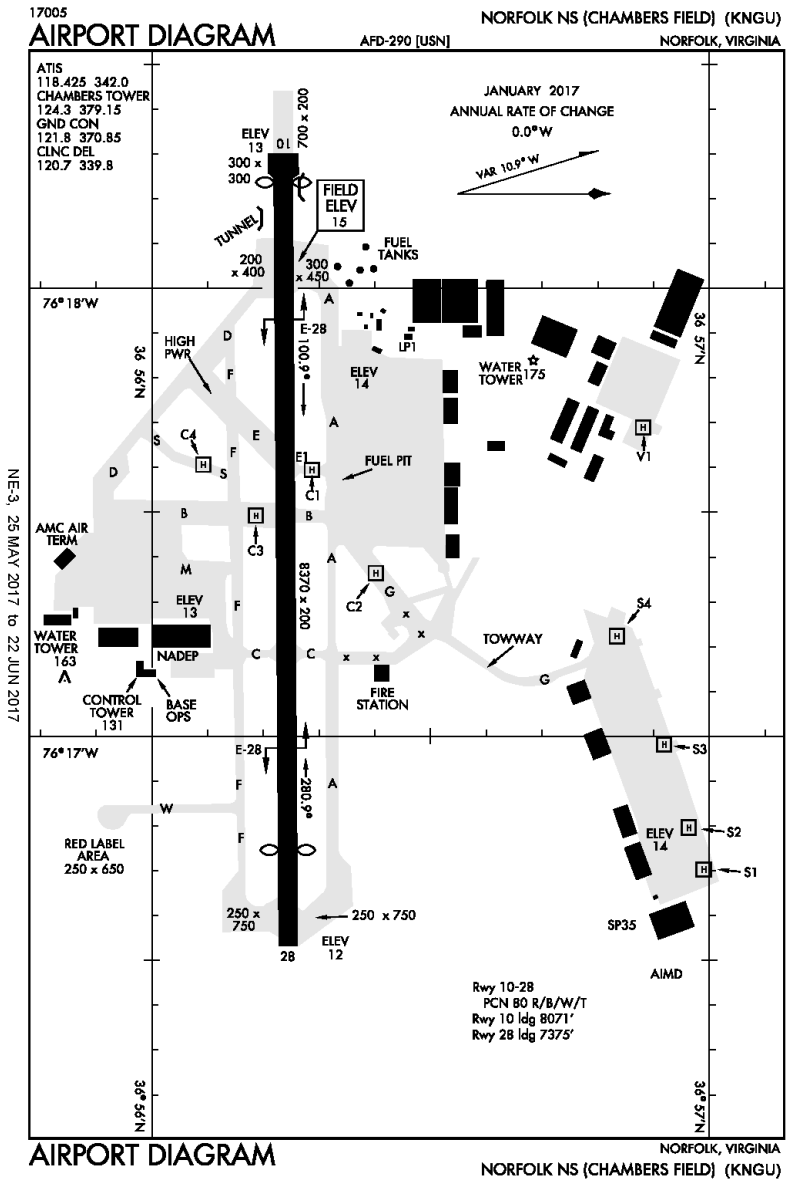
RADAR INSTRUMENT APPROACH MINIMUMS

RADAR MINS

17033

N2

Example KNGU Airport



FFS CATII/III PRM/PAR KSEA

Preflight

- Abbreviated procedures
- Program FMC, set trim, airspeeds, and MCP
- Instructor will configure overhead panel

Pushback/Start

- Pushback/Start procedure
- Fast start accomplished by instructor

Taxi Out

- Taxi Out procedure
- During taxi demonstrate differences in forward visibilities, 150m / 500', 300m / 1000', 350m / 1200' and then back to takeoff Mins (500)
- Demonstrate the use of Stop Bars, Centerline Lights, Geographic Position Markings and Directional Low Visibility Taxi Routes.

Takeoff – Normal Procedures

- Instrument, Crosswind

Instructors Note
With 2 Captains or 2 First Officers crew pairing, both trainees must accomplish CAT-II / CAT-III approach requirements from their respective crew position.

Approach and Landing

ILS CAT III approach – Full Stop

- Position freeze at 100' and 50'
- Approach callouts
- Visibility demonstration
- RVR 1200, 1000, 600, and 300
- Discuss seat height and cutoff angle
- Mode failures - Land 2 vs. Land 3
- FMAs
- Exiting runway and low visibility taxi charts
- A/P disconnect after landing

FFS CATII/III PRM/PAR KSFO

Reposition to 8 – 12nm final

ENG Inop ILS CAT II Approach – MAP

- Engine inoperative
- Flight Mode Annunciations
- Autopilot missed approach

Reposition to 8 – 12nm final

ENG Inop ILS CAT II Approach – Full Stop

- Flight director information
- Use of autothrottles
- Flight Mode Annunciations
- Select one Mode Failure
    - Auto Speedbrake, Autobrakes, Reverser, Rollout

Instructors Note
Aircraft may be repositioned for additional
CAT II / CAT III approaches as necessary.

Instructors Note
With 2 Captains or 2 First Officers crew pairing, both trainees must accomplish PRM breakout procedure from
Their respective crew positions.

Instructors Note

KSFO PRM Attention All Users Page and the ILS PRM 28L charts are located on JEPP FD PRO. They are located in the Company Tab, The LDA chart is no longer available.

Reposition to 8 – 12nm final

ILS /PRM approach

- Breakout maneuver
- Crew duties
- Aircraft configuration

All Simulators – Select the ILS approach for 28L and conduct the PRM approach.

"TRAFFIC ALERT" *turn right immediately heading 010, climb and maintain 4,000".*

Instructors Note
Aircraft may be repositioned for additional ILS/PRM Breakouts as necessary.

FFS CATII/III PRM/PAR KSFO

Reposition to 8 – 12nm final

PAR / GCA Approach

- CAT-I minimums
- Normal procedures
- PAR approach
- Use of Automation
- Flight director information
- Flight Mode Annunciations
  - V/S and HDG SEL
- Complete full stop landing

Requirements

- PAR approach
- Flown by Captain

- Crew should complete final landing configuration and checklist.
- Captain sets the DA in the MCP Altitude window.
- ATC Communications should take place at least every 1 minute when being vectored to Final Approach, and at least every 5 seconds when being vectored on Final Approach..

"

Begin descent".

- *Clearing runway*

- Captain selects the V/S and sets descent rate
- Approx 3 degree glide path;

Grd Spd	V/S
140 -	750fpm
160 -	850fpm
180	1000fpm

- Controller provides course and glide path guidance as necessary;

"Slightly / well - above / below glide path"
"Slightly / well - left / right of course"
"Slightly / well - left / right of course and holding / correcting"
"Going Above / Below Glide path"
"Going Above / Below Glide path and Holding / Correcting"
"Going Above / Below Glide path and coming down / up"
"On Glide path / On Course
"Decision Height"
"Over Approach Lights"

- Crew advises Runway in Sight.
- Captain Disconnects A/P
- Controller continues to provide advisory course and glide path information until over the landing threshold at which time the controller will advise;

Over the landing threshold".

- Captain disconnects the A/T and lands.

Note:To stop the controller from issuing instructions, call runway in sight.

Right Seat Takeoff and Landing

- With remaining simulator time, perform right seat takeoffs and manual landings.
    - It is required to document Captains right seat takeoff and landing.
    - One takeoff shall include an engine failure after V1 and prior to V2.
    - One approach shall be a manually flown 3 Engine ILS to minimums.